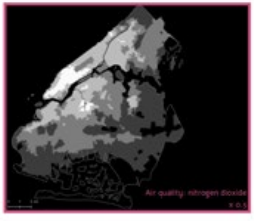
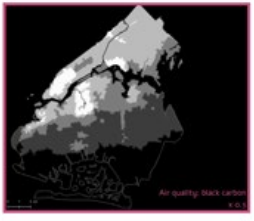
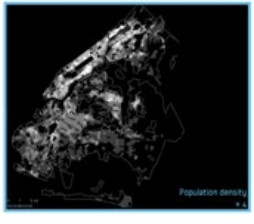
Poetics & GIS Overlays: Black Ecological Futurity & Streetscapes, NYC

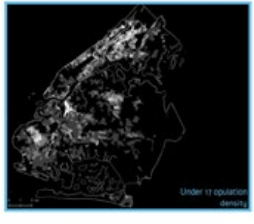
a working landscape architecture graduate thesis

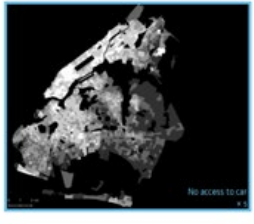
GIS Overlays

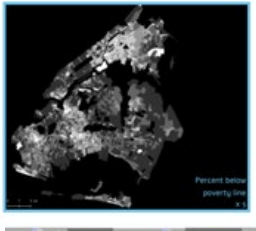




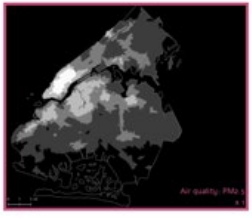


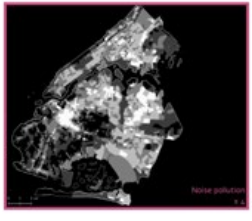


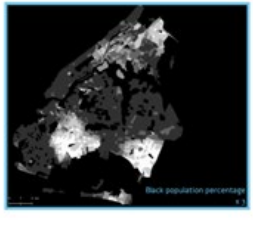


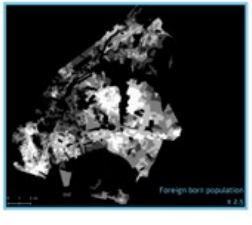


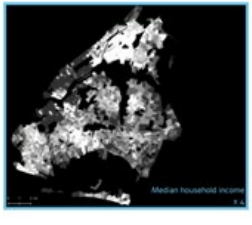
See OVERLAYING* on the next page for more context on how these maps were used, and what the number (weight) next to them means.

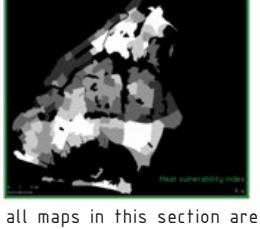












all maps in this section are made by author, unless stated otherwise

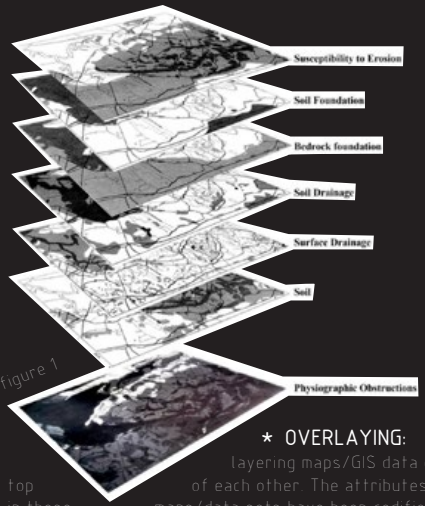


figure 1

to "design with nature," and make envrironmentally sound land use/land development decisions, ian mccargh published his book "design with nature" in 1969.

in the book, he shows how incorporating/OVERLAYING* various landscape characteristics (such as soil drainage, soil type, bedrock formation, etc.) and ecology can result in landscapes that create a sense of space for the inhabitants, and are resilient againt forces of nature.**

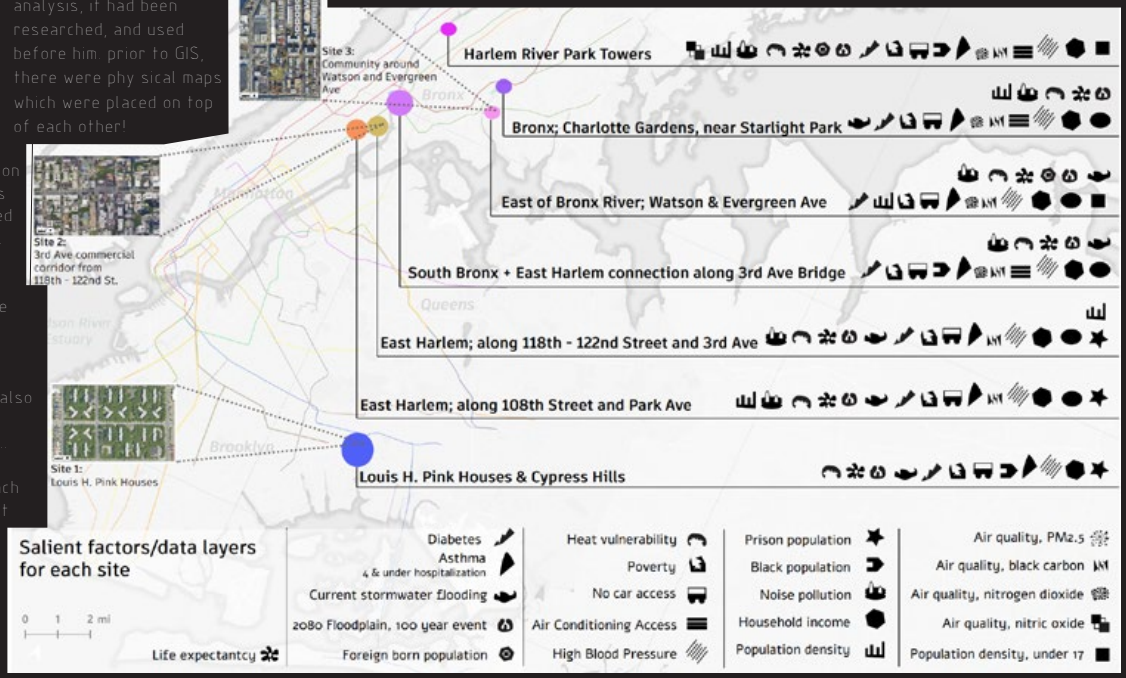
** although ian mcharg popularized this overlay method, or multicriteria analysis, it had been researched, and used before him. prior to GIS, there were phy sical maps which were placed on top of each other!

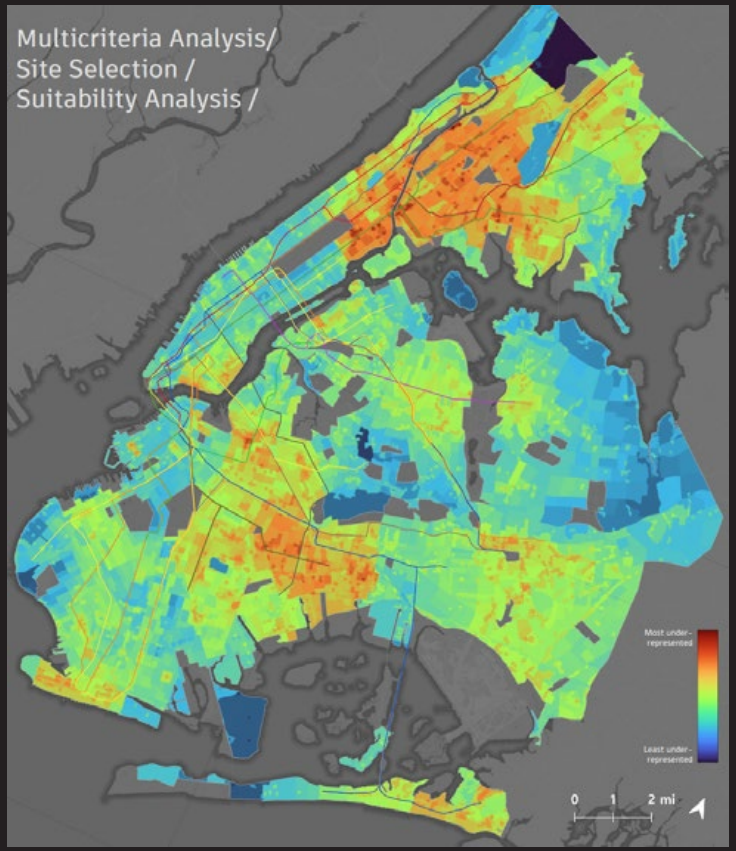
The image on the far left shows various landscape characterisitcs that were used in original overlay method. The map below shows the various datasets I used in my analysis, which includes demographic and health datasets. Current usage and complilation of such data is called an INDEX.

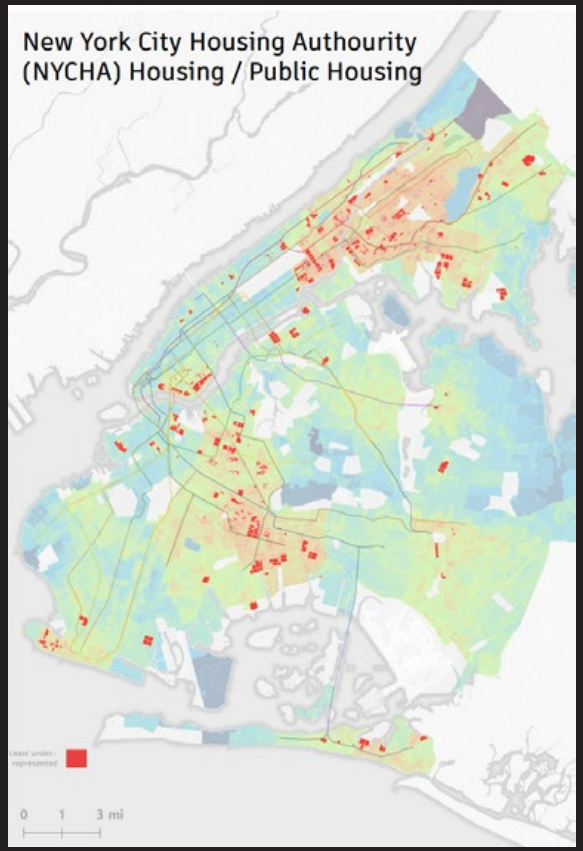
* OVERLAYING: layering maps/GIS data on top of each other. The attributes in these maps/data sets have been codified from 0-5, with 0 as unsuitable and 5 as suitable. for example, if deciding where to install additional drainage, locations with nuisance flooding might be codified as more suitable (3-5), than locations where there is none (0).

visually, attributes that are given a value of 0 are darkest and value of 5 are lightest. attributes can also be given an value of 1, 2, 3, and 4 on the suitability scale, and these values are seen as shades of grey.

all the codified datasets/maps are laid on top of each other, and suitability of each dataset is combined. at this stage, certain datasets can also be "weighed" more than others. for example, when deciding where to install drainage, nuisance flooding dataset can weigh 2x, while projected future storm water surge dataset could weight 0.5x







ASIDE: New York City is seeing the privatization of its public housing developments.

Due to decades of neglect and lack of maintenance, many buildings have fallen into dangerous disrepair. NYCHA/ neoliberals proclaim that working with private developers, and opening some of apartments to market forces, is the only way out of this predicament.

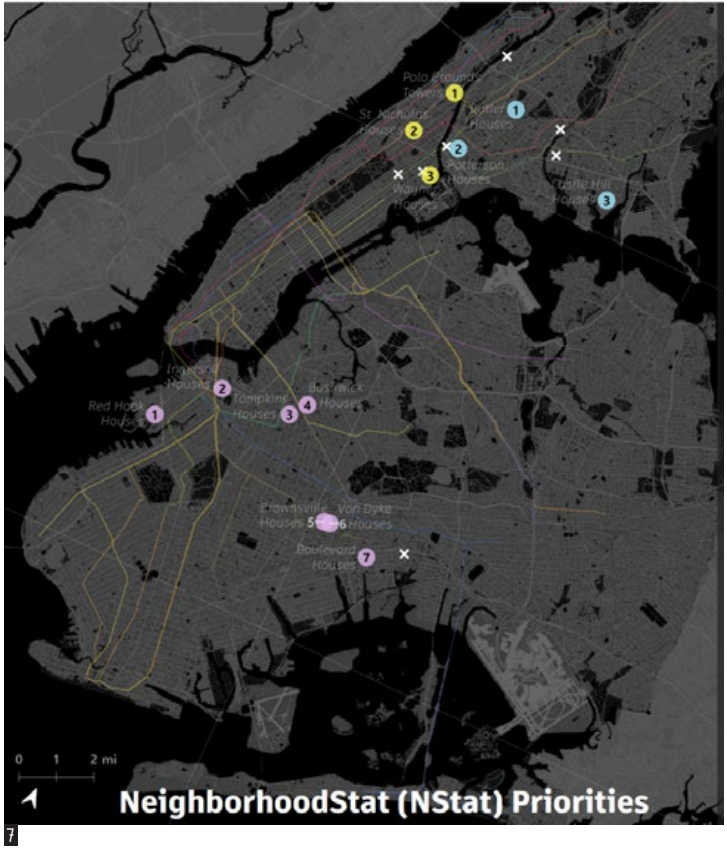
Privatization through strategic abandonment is a strategy. Just like burning buildings for insurance payouts.

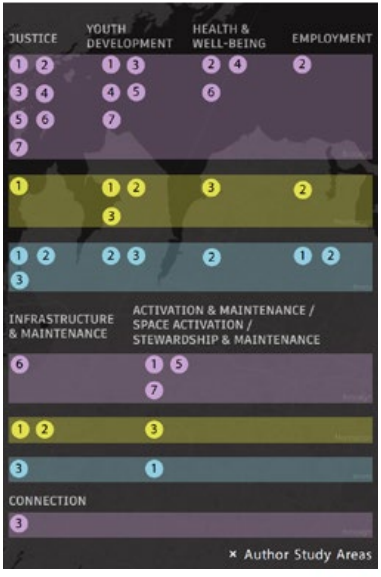
Visions for ways to convert these developments into community land trusts is a worthwile dream.

The outcome of overlay method/multicriteria analysis I used for this thesis is the map on top-left (above). Red/orange locations are those I found to be most under-served and under-resourced.

This outcome overlapped significantly with NYCHA housing. This shaped my research and approach going forward. Black ecology became a grounding scholarship on how to conceptualize futurity and streetscape.

[Lack of] Community Engagement: due to lack of funds, and firm belief in compensating residents for their expertise, I don't directly speak with residents of communities highlighted in this analysis.





On this spread. are takeaways from a program developed by the Mayor's Office of Criminal Justice: Neighborhood Stat (NStat),

I use these reports because the NStat project works with residents of public housing (NYCHA) across the city. The priorities were established with community members, and offers a citywide look at some shared concerns, wants, and desires

GIS overlays

Olalekan Jeyfious' World-Making

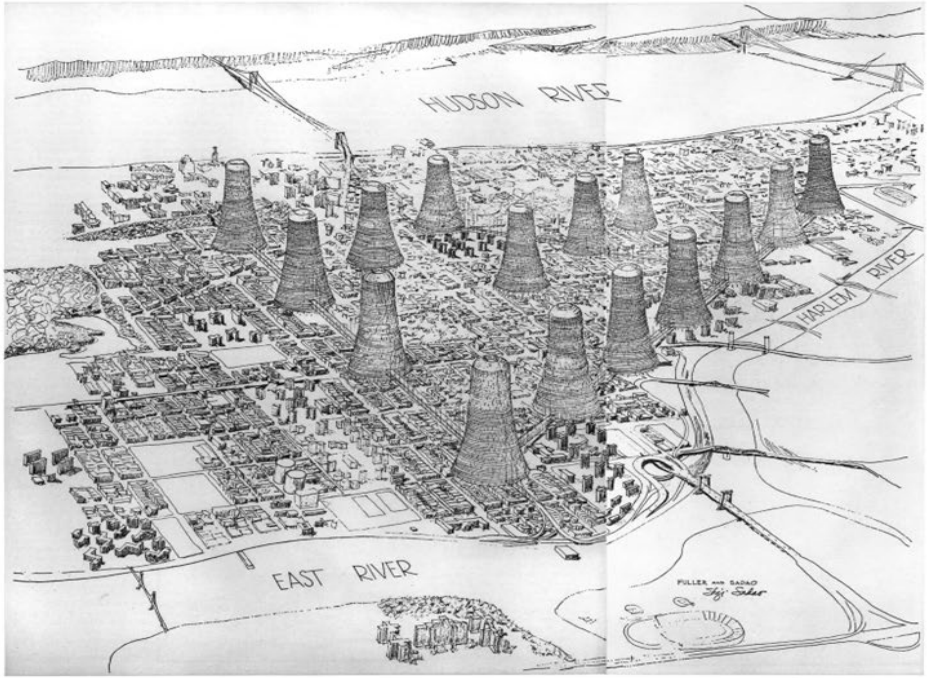
Jeyfious projects show futurities where many priorities, highlighted in NStat repoort, are addressed such as "activation

and maintenance," "youth development," and "infrastructure and maintenance." Jeyfious' images often center the youth and alternate education systems, where the city and environment is a place for learning. Morever, there are no lawns, which is a common sight in NYCHA developments, and the outdoors are teeming with infrastructure which undoubtedly needs stewardship and maintenance.

The health benefits of outdoor, green spaces have been recorded, especially since the pandemic, and one can assume that with so much green space, that NStat priority is being actively addressed.

Jeyfious' work is laden with the poetics of black experience

This publication in Esquire magazine initially failed to credit June Jordan as an author, and alters the title from "Skyrise for Harlem" to "Instant Slum Clearance." Slum clearance was a racialized practice of the State working with developers to displace black residents from their homes.



Instant Slum Clearance

by June Meyer

R. Buckminster Fuller designs a total solution to an American dilemma: here, for instance, is how it would work for Harlem

by: JUNE JORDAN & R. B. Fuller

Harlem is life dying inside a closet, an excrescence beginning where a green park ends, a self-perpetuating disintegration of walls, ceilings, doorways, lives. It is also, of course, a political embarrassment for which no political solution is adequate. A housing project planted in the middle of a slum is not an answer. Harlem has been much lamented, but those statistics may be less familiar than others:

1. A typical Harlem child will score lower on an I.Q. test in the sixth grade than he scored three years earlier in the third grade.
2. Nine-tenths of its housing units are more than thirty years old.
3. Half the youngsters live with one parent or with none.
4. Harlem has a population of a quarter of a million, but it doesn't contain a high school.
5. Traffic deaths for Harlem youths appreciably exceed the rate plaguing the whole island of Manhattan.

Skyrise for Harlem is a proposal to rescue a quarter million lives by completely transforming their environment. New Harlem will encompass a half million people by removing old limits in exchange for natural boundaries. Harlem will widen from river to river across the island. Its new space will accommodate an additional quarter million residents—everyone willing to participate in the integrated transformation of a ghetto.

Skyrise for Harlem can be completed in thirty-six months. The first year will be spent in what R. Buckminster Fuller describes as "tooling up": organizing the mass production of structural parts and utility units, including all basic furniture.

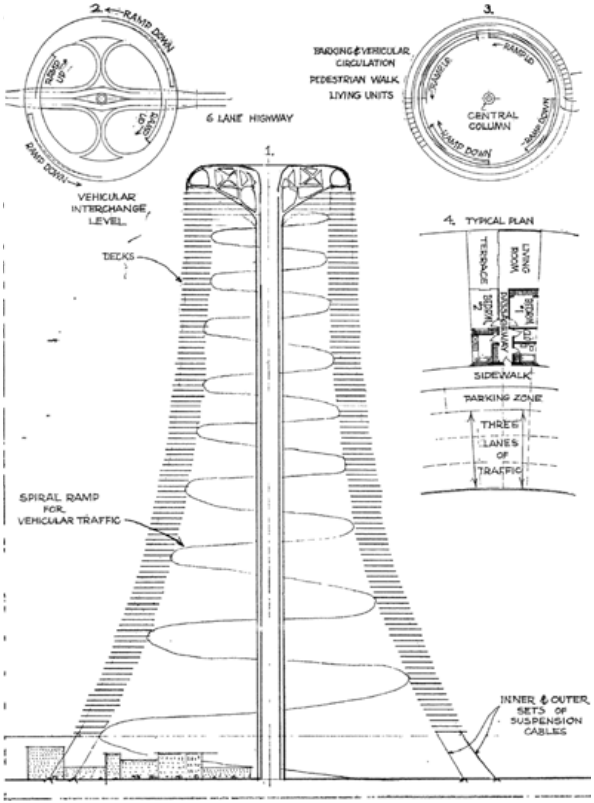
Redevelopment generally means the removal of slum residents

Before and After, in the photograph below, Harlem today. In Buckminster Fuller's rendering at left, the same area as it would be re-created. Central Park ends at the extreme left, the George Washington Bridge is at the upper right. Traffic will flow over a new bridge to be built at 125th Street into the system of elevated, interconnected highways connecting all fifteen towers. These highways will handle through traffic in all directions and permit exit into the buildings themselves.

"No one will move anywhere but up. New Harlem will be supported by columns driven into the back-yards of the slum, and once the elevated replacement is complete and inhabited, the lower depths will be cleared for roadways and park space.".
- quoted in article to left

Jordan was actuely aware of how the built environment impacts the quality of life, and in response to slum clearance and racial terror (1964 riots), she worked with Fuller to imagine another reality for the black + brown residents of Harlem. Harlem, especially East Harlem, is one of the most under-resourced communities in NYC.

"I would think that this new reality of Harlem should immediately reassure its residents that control of the quality of survival is possible and that every life is valuable... I am much heartened by your insistence on the invention of a physical device rather than efforts of social reform. I also believe that the architecture of experience deeply determines an incalculable number and variety of habits—i.e., the nature of quotidian existence." – June Jordan, Letter to R. Buckminster Fuller (1964)

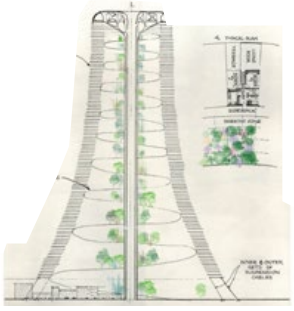


one of the datasets used in the analysis is no car access.

neighborhoods which had no/limited car access were ranked as very suitable for reimagining the streetscape. the streets are the largest, continuous public asset, and decentering cars would open vast amounts of space for other ways of relating and living.

with that consideration, i experiment with jordan and fuller's skyrise, by replacing traffic lanes with trees/shurbs/herbaceous plantings.. these are placeholding elements to start ant think about other ways that the space could be used, including playgrounds, community spaces, etc,.

furthermore, micromobility is starting to redefine how residents move across the city, and envisioning the ways that residents would move within and outisde the structure could encourage a lively discussion on how (un)-contained the skyrise is; and how it can connect to the outside.



left: a quick addition, by me, to jordan and fuller's design, which replaces car/traffic lanes with plantings

BLACK ECOLOGY

ways of relating
prefiguring mutual aid
centrality of the quotidian

"Flinders asks readers to think of the values of belonging as "points on a circle, windows onto a single reality." Listing the characteristics of the culture of belonging, Flinders explains that: "Fundamental to the culture of belonging is a strong sense of reciprocal connection to the land where one lives, empathic relationship to animals, self restraint, custodial conservatism, deliberateness, balance, expressiveness, generosity, egalitarianism, mutuality affinity for alternative modes of knowing, playfulness, inclusiveness, nonviolent conflict resolution and openness to spirit."
- bell hooks, "belonging", page 217

"The poetics of place are the veritable archive of consciousness, subaltern visions, insurgent spatial practices and knowledge that work to build not only material Black spaces, but socio-symbolic Black worlds. By poetics, Black geographies scholars speak of the stories, recipes, food practices, musical traditions, poetry, and grounded theories that express and instantiate Black people's understandings and experiences of space and as well as their place-making and place-taking practices (McKittrick 2006; McKittrick and Woods 2007; Woods 1998)"
- Moulton, Alex A. and Inge Salo, "Black Geographies and Black Ecologies as Insurgent Ecocriticism"

"We All We Got Urban Black Ecologies of Care and Mutual Aid"
by: Ashanté M. Reese and Symone A. Johnson

"When the pervasiveness of anti-Blackness meets climate-related catastrophes that the state refuses to plan for or mitigate, it amounts to planned abandonment, or the state's gambling with how many (Black) lives are expendable before needing to intervene or make drastic changes to prevent or reduce loss in the future (Woods 2017; see also Ananya Roy's 2019 work on racial banishment). Indeed, "communities feel disaster impacts in racialized terms. Anti-Blackness shapes communities' exposure to environmental hazards and disasters" (Purdum et al. 2021: 3)" -page 37

"We therefore amend Ahmed's (2004) contention to say that it is in fact white feelings that do things. White feelings build infrastructures of violence while at the same time Black feelings—in particular, otherwise feelings of joy, hope, love, and mutuality made possible by practicing interdependence through care—are "illegible", "unintelligible", "unthinkable" in the white imaginary (Palmer 2017; Rankine 2014; Trouillot 1995). Due to their unintelligibility, Black networks of care and mutual aid provide viable infrastructural material for the establishment of maroon communities inside geographies of violence—hidden in plain sight (Roane 2018). As Celeste Winston argues, these maroon geographies are made up of both physical sites established by those escaping enslavement and more contemporary examples such as "spaces produced through continued Black struggles around policing, incarceration, housing insecurity, unequal food access, environmental racism, and other overlapping forms of racial violence" (2021: 2187). As a result, these maroon geographies, "advance alternative ways of understanding and producing space against and outside of these structures of racial violence rooted in the history and legacies of slavery" (Winston 2021: 2187)" - page 32

Hurricane Katrina in 2005 and Winter Storm Uri in 2021, in New Orleans and Austin, TX respectively, saw state abandonment in different ways. In Austin, TX areas with higher black resident concentration saw "pre-cautionary" energy blackouts, and Hurricane Katrina saw black and brown communities declared inhabitable even as houseless, poor, and working class residents used them for shelter. The lack of foresight and preparation for these "natural" disasters points to a pattern of state abandonment (or privatization, or both). Black Ecology shows us how black communities resilience is embedded in its relationships and affective communities, which are "unintelligible" to hegemonic systems and populations. Therefore, anyone who sincerely wants to work with the communities has worlds of knowledge and relating to learn and unlearn.

poetics, black ecology, futurity

science fiction : speculative fiction // climate fiction

everything for everyone, an oral history of new york commune 2052-2072
by: m.e. o' brien and eman abdelhadi (2022)

new york: 2140
by: kim stanley robinson (2018)

"Desperate to reignite national frevor and unity as well as restart the economy, the United State began an extended invasion of Iran in 2040. The war lasted eight years, wreaking a massive and devastating toll on Iranian life. It also destroyed the last of the already fragile legitimacy of the US ruling class… in 2045, by the estimates of researchers in US military intelligence, more than eight thousand commanding officers in the US Armed Forces were murdered by their subordinates. By the end of the war, the US military was in chaotic disarray and open rebellion… in 2056, the US federal government declined military aid to the beleaguered NYPD, leading to the defeat of the police in the following weeks." - pages 11-12

US attacked Iran this year, in 2026, 14 years before the one in this piece of speculative fiction. Using their timeline as a guiding cue, I use the year 2050 to envision an abolitionist reality in NYC.

"…on the open-walled farm floor of the old Met Life tower, from which vantage point lower Manhattan lies flooded below them like a super-Venice, majestic, watery, superb. Their town.
…
Yeah but this is a big city! Lots of nooks and crannies, lots of dark pools, the underwater economy and all. We could dive in and disappear." - page 6

"The assembly provided the main form of decision making in these new insurrections. Assemblies initially referred to nightly or weekly gathering on street corners during riots and other urban uprisings" - page 14

public places, esp. the streets are a critical place for praxis of radical, direct democracy. throughout the book, we can see how they play a range of roles in the everyday lives of the people. from acting as market places to outdoor venues for nightlife activities. in BLACK ECOLOGY scholarship, we see how people come together for mutual aid, collective grief, festivals, etc,

The official move away from a carceral, white-supremacist, hetero-patricarchial imperalist system doesn't result in an stable alternative overnight. However kernels of what we practice now can be a rich, fertile ground for developing/experimenting towards a futurity that's grounded in a radically inclusive quotidian.

"I didn't think there were skybridges between here and there." "Nothing direct," Gen explained, "but you can take the one from here to Bellevue, and then go downstairs and cross diagonally and then head west on the Twenty-third Skyline.

"Gen led her to the elevator in Bellevue's northwest annex that would take them down to the skybridge that ran west from building to building on the north side of Twenty-third. Most skybridges still ran either north-south or east-west, forcing what Gen called knight moves. Recently some new higher skybridges made bishop moves," - pages 10-11

"We're squeeze dry. We've been paying a fraction of what things really cost to make, but meanwhile the planet, and the workers who made the stuff, take the unpaid costs right in the teeth." - page 4

there is a financial trader in the book who bets on sea level rise and housing stock. the politico-economic reality is similar to present day.

in new york 2140, the sea level rose around 50 feet, and lower manhattan has become like venice. throughout the book, there are details on how the charers manuever their boat through this area. this watery future also leads to skybridges and the first 4-5 floors turning into boat docks.

these two books present very different futures. one future is abolitionist/decentralized, while the other focuses on the logistics of NYC that's underwater, the former is also impacted by the climate/ecological crises, but the focus is on the relationships amongst humans, other species, and earth tiself == the relationality is radically different than present day, whereas NY2140, has relationships and politico-economic dynamics we would recognize now, but changes to the physical environment are described in great detail. these are the respective strengths of the books, which helps to conceptualize a narrative regarding the future.

futurity & streetscapes, NYC

References (in order of presentation)

pages 1-2:

data from 2020 US census and nyc open data

page 3:

figure 1 - widely attributed to ian mcharg, "design with nature" (1969); however, source unconfirmed

pages 7-8:

https://map.cityofnewyork.us/neighborhood-stat/
neighborhoodstat (NStat) website. information about each community can be found under the "COMMUNITIES" tab/drop down menu. once a community is selected, the priorities information is avaibale in the central tab.

page 9:

olalekan jeyfious' instagram @kidcadaver
specific project posts (carousels): 1, 2, 3

page 10:

olalekan jeyfious' "[TFN] the frozen neighborhoods" exhibit at MoMa: RECONSTRUCTIONS: ARCHITECTURE AND BLACKNESS IN AMERICA (2021)

pages 11 - 14:

"skyrise for harlem or "instant slum clearance" by JUNE JORDAN and R. BUCKMINSTER FULLER in esquire magazine (1965)

Unreferenced references (incomplete)

ARTICLES

"june jordan's politcal theory of redesign" by LISA BEARD, american political science review (2025)
"place, emotion, and environmental justice in harlem: june jordan and buckminster fuller's 1965 "architextual" collaboration" by CHERYL J. FISH, discourse: race, environment, and representation (2007)
"push & pull" by BRADFORD MCKEE in landscape architecture magazine (2023)
"the value of land: how community land trusts maintain housing affordability" by OKSANA MIRONOVA, urban omnibus (2014)

BOOKS

"abolition geographies: essays towards liberation" by RUTH WILSON GILMORE (2022)
"ambiguous territory: architecture, landscape, and the postnatural" edited by CATHRYN DWYRE, CHRIS PERRY, DAVID SALOMON, and KATHY VELIKOV (2017)
"infrastructure, well-being, and the measurement of happiness" edited by HODA MAHMOUDI, JENNY ROE, and KATE SEAMAN (2023)
"reconstructions: architecture and blackness in america" edited by SEAN ANDERSON and MABEL O. WILSON (2021)
"speculative futures: design approaches to navigate change, foster resilience, and co-create the cities we need" by JOHANNA HOFFMAN (2022)
"waterproofing new york" edited by DENISE HOFFMAN BRANDT, and CATHERINE SEAVITT NORDENSON (2015)

JOURNALS & REPORTS

"global black ecologies" edition of environment and society, edited by JUSTIN HOSBEY, HILDA LLORENS, and J. T. ROANE; volume 13 (2022): issue 1 (2022)
"making space: youth-centered spatial futures in nycha communities"by URBAN DESIGN FORUM FOREFRONT FELLOWS, FREE TO GROW (2024)

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recommendations

constructive criticism

deeply appreciated and valued <3

peace